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Individual Differences in Makerspace Learning: Pose-Based Student Profiles and Their Relation to Self-Efficacy

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Corresponding Author:	Bertrand Schneider Harvard University Cambridge, MA UNITED STATES
First Author:	Jiawei Zhang
Order of Authors:	Jiawei Zhang
	Jack Basmaci
	Bertrand Schneider
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Cover letter

Professor Bertrand Schneider
Longfellow Hall, office 333
School of Education
Harvard University
13 Appian Way Cambridge, MA 02138
bertrand_schneider@gse.harvard.edu

Harvard
Graduate School
of Education

August 22nd 2026,

Dear Editors,

Please consider our manuscript, titled **"Exploring Individual Differences in Makerspace Learning: A Multimodal Analysis of Self-Efficacy, Social Interaction, and Learning Trajectories,"** for publication as a research article.

This manuscript examines how individual differences in psychological constructs (specifically self-efficacy and sense of belonging) interact with social behaviors to shape learning outcomes in a university makerspace environment. Leveraging a large-scale multimodal learning analytics (MMLA) dataset, we analyze high-frequency behavioral data, including joint visual attention (JVA) as a proxy for social interaction, to understand the varied "journeys" students take in open-ended, constructionist settings. While prior research has often focused on laboratory-based experiments, our study provides a rare look at these dynamics within an ecological, technology-rich environment, offering insights into how personalized support and affective feedback can be scaled to meet diverse student needs.

Key contributions of this work include:

- **Methodological Innovation:** The use of motion sensors and computer vision to quantify social interactions (JVA) at scale in physical learning spaces.
- **Empirical Findings:** Evidence connecting social-emotional growth and identity formation to technical competency development.
- **Practical Implications:** Guidance for designing interventions, such as AI-generated personalized feedback, that account for individual differences in student confidence and community belonging.

This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. All authors have approved the submission.

Thank you for your time and for considering our work for publication. We look forward to your response.

Sincerely,



Title page with author details

Individual Differences in Makerspace Learning: Pose-Based Student Profiles and Their Relation to Self-Efficacy

Jiawei Zhang, Harvard University, jiaweizhang@gse.harvard.edu

Jack Basmaci, Harvard University, jab5200@g.harvard.edu

Bertrand Schneider, Harvard University, bertrand_schneider@gse.harvard.edu

Corresponding author: Bertrand Schneider, Harvard University, Graduate School of Education

13 Appian Way, Longfellow Hall 333, Cambridge, MA 02138,

bertrand_schneider@gse.harvard.edu

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Individual Differences in Makerspace Learning: Exploring Pose-Based Student Profiles and Their Relation to Self-Efficacy

Abstract

Makerspaces can support powerful learning by positioning students as active designers, but students do not all participate the same way. This study examines whether pose-based behavioral data can profile these differences and connect them to self-efficacy. Using pose data from a semester-long digital fabrication course, we derived student profiles and Ordered Network Analysis (ONA) features organized around Bandura's self-efficacy theory: mastery, vicarious experience, social support, and affective regulation. Results revealed two stable dimensions: an affective pathway distinguishing stress-regulated from stress-burdened students, and a behavioral pathway distinguishing deep from surface engagers. Combining these profiles related more strongly to self-efficacy than either alone, depending on students' sense of belonging. ONA showed deep engagers had stronger behavioral connections related to mastery, vicarious learning, and social persuasion. These findings suggest pose-based profiles can complement surveys and observations by revealing the embodied pathways through which students participate in makerspace learning and develop confidence as makers.

Educational Relevance and Implications Statement

This study shows that everyday video of students working in a makerspace can reveal meaningful differences in how they engage, differences that are not obvious from grades or one-time surveys alone. Some students engage deeply and confidently, others engage just as much but experience it as stressful, and these different pathways relate to how capable students feel as makers. This suggests that instructors cannot rely on activity level alone to judge how a student is doing, and that students who look busy may still need support if they are stressed or working in isolation. For practice, this points to using lightweight behavioral signals, alongside conversation and observation, to help instructors notice which students may benefit from encouragement, structured peer support, or a stronger sense of belonging in the makerspace.

1. Introduction

Constructionist theories of learning suggest that "making" can transform students' mindsets by positioning them as active builders of knowledge who learn through designing, testing, revising, and sharing personally meaningful artifacts (Papert, 1980; Blikstein, 2013). Supporting these students, however, is unusually demanding for makerspace facilitators because students do not all move through makerspace learning in the same way. Some students quickly become deeply engaged with tools and projects, while others remain more peripheral, move in and out of participation, or experience making as stressful even when they are making progress. When everything works well, students develop what is called a "maker mindset" (Dougherty, 2013): a willingness to experiment, persist through failure, learn from peers, become legitimate participants in a community of makers, and see themselves as capable of designing and building things (Blikstein, 2013).

However, understanding how students develop a maker mindset is difficult in practice, especially when many students are working on different projects, using different tools, and encountering different technical (and emotional) challenges. For this reason, makerspace facilitators and researchers tend to rely on what is easiest to observe: completed artifacts, self-reported reflections, surveys, or moments when students explicitly ask for help. Ideally, these sources help us understand how students learn in makerspaces - but they can miss important parts of the learning process, especially the moment-to-moment behavioral patterns through which students engage with tools, peers, and the physical space.

In this paper, we explore whether pose-based behavioral data can provide an additional lens for studying individual differences in makerspace learning. We use video-derived pose data to characterize how students physically move, pause, observe, collaborate, seek help, and interact with machines over the course of a semester. This way, we can examine not only whether students report changes in self-efficacy, but also whether distinct behavioral profiles are associated with different patterns of engagement. The goal is not to replace surveys, reflections, or instructor observations, but to complement them with a more continuous and fine-grained representation of students' activity in the makerspace.

We expect these behavioral profiles to be related to students' self-efficacy because self-efficacy is central to how students engage with open-ended tasks (Bandura, 1977). In a makerspace, students are often asked to work with unfamiliar tools, troubleshoot ambiguous

problems, and persist through cycles of failure and revision. Under these conditions, students' belief that they can solve problems, recover from setbacks, and continue improving is likely to shape whether they take risks, seek help, iterate on their designs, and remain engaged with their projects. Pose-based profiles may therefore help us identify different pathways through which students enact, struggle with, or develop this belief over time.

Additionally, we focus on the affective side of makerspace participation. Stress, enjoyment, and time spent working are part of how students experience open-ended making. A student who spends many hours in the makerspace may be deeply engaged, overwhelmed, or both. A student who appears less active may be disengaged, observing strategically, or regulating their participation in ways that are difficult to capture from surveys alone. By connecting behavioral profiles with affective reports and changes in self-efficacy, we can better understand how different students inhabit the same learning environment.

We start by reviewing the literature on individual differences in makerspace learning, self-efficacy, and behavioral trace methods. We then describe how we collected and processed pose data from a semester-long digital fabrication course, derived student profiles, and constructed Ordered Network Analysis features to represent the temporal organization of student behavior. We test whether these profiles differ in Bandura-aligned behavioral pillars and examine how these patterns evolve across the semester. We discuss what these results suggest for studying makerspace learning and for designing more responsive support for students with different learning trajectories.

2. Literature review

Constructionist theories of learning suggest that makerspaces can be powerful learning environments because they position students as active designers of personally meaningful artifacts (Papert, 1980). In makerspaces, students learn by building, testing, debugging, revising, and sharing things they care about in a community of practice. This kind of learning can be especially generative because it connects conceptual understanding with material activity. Students encounter ideas through tools, materials, prototypes, failures, and conversations with peers. When things go well, they begin to see themselves not only as students completing assignments, but as people who can design, make, troubleshoot, and contribute to a community of makers (Martin, 2015).

2.1 Individual differences in maker-centered learning

Makerspace learning is difficult to study because it is open-ended, social, embodied, and highly variable across students. In a single course session, one student may be laser cutting, another may be debugging code, another may be asking a peer for help, and another may be stepping back to observe how a machine works. These differences matter. Makerspace learning is not only about whether students complete a final artifact, but also about how they move through the space, how they interact with tools, how they respond to uncertainty, how they seek help, and how they regulate stress and frustration (Sheridan et al., 2014). For this reason, researchers have argued that makerspaces should be understood as complex learning environments where participation involves technical practice, social interaction, identity development, and affective experience (Blikstein, 2013).

One important challenge is that students do not experience makerspaces in the same way. Some students enter with prior technical experience, while others are using fabrication tools for the first time. Some students quickly become comfortable experimenting with machines and materials, while others may experience the same environment as intimidating or stressful (Melo & Rodney, 2023). These differences can shape how students participate. A student who feels confident may be more willing to take risks, troubleshoot independently, or iterate on a design. A student who feels uncertain may spend more time observing others, seeking help, or avoiding particular tools. These patterns are not necessarily good or bad by themselves. Observing, collaborating, and asking for help can all be productive parts of learning. However, the way these behaviors unfold over time may reveal important differences in how students engage with makerspace learning.

For example, in a university makerspace study, Andrews et al. (2021) found overall gains in design self-efficacy, innovation self-efficacy, technology self-efficacy, makerspace belonging, and engineering-community belonging, but also reported meaningful differences across student characteristics such as year in program, gender, and race. Similarly, Hilton et al. (2020) examined engineering design self-efficacy among makerspace participants across three universities and showed that makerspace involvement and self-efficacy are intertwined with demographic variation. At younger ages, Vongkulluksn et al. (2018) found that students' self-efficacy and situational interest in a makerspace course changed over time and were related to achievement emotions such as excitement and frustration. Together, these studies suggest that makerspaces are not experienced as uniformly empowering or motivating environments. Instead,

students' prior experiences, identities, affective responses, and patterns of participation shape how they engage with making and what they take away from it.

2.2 Self-efficacy in makerspaces

Self-efficacy provides one (but not the only) useful framework for understanding these differences. Bandura (1997) defines self-efficacy as people's belief in their ability to organize and carry out actions needed to accomplish particular goals. In open-ended learning environments, self-efficacy is especially important because students often need to persist through ambiguous problems, repeated failures, and uncertain outcomes. In a makerspace, students may need to learn unfamiliar software, operate tools that seem intimidating, debug physical and digital systems, and revise their designs after things break or fail. Under these conditions, students' beliefs about their ability to solve problems are likely to shape how they respond to challenges.

Bandura's theory also explains where self-efficacy comes from. Students develop self-efficacy through mastery experiences, vicarious experiences, social persuasion, and affective or physiological states (Bandura, 1997). These four sources map naturally onto makerspace learning. Mastery experiences occur when students successfully use a tool, solve a design problem, or make progress after repeated attempts. Vicarious experiences occur when students learn by watching peers, teaching fellows, or instructors use a machine, debug a prototype, or approach a design challenge. Social persuasion occurs when students receive encouragement, guidance, or feedback from others that helps them believe they can continue. Affective states are also central: stress, frustration, excitement, and enjoyment can shape how students interpret their own competence.

This framework is useful because it connects observable behavior to psychological development. If students' self-efficacy is shaped by mastery, observation, social support, and affective regulation, then makerspace behavior may contain traces of these processes. For example, repeated interaction with machines may reflect opportunities for mastery. Moving from observing a machine to using it may suggest a transition from vicarious learning into hands-on practice. Help-seeking during tool use may reflect social support or regulation during moments of difficulty. Time spent working alone after tool interaction may reflect reflection, self-regulation, or disengagement depending on the broader behavioral context. The key point is that

self-efficacy is not only measured through surveys at the beginning and end of a course. It may also be visible, indirectly, in how students participate over time.

2.3 Using multimodal learning analytics to capture learning in makerspaces

Studying these patterns requires methods that can capture the fine-grained and temporal nature of makerspace activity. Traditional measures, such as surveys, interviews, or final project assessments, provide important information, but they often miss how learning unfolds moment by moment. A student's average enjoyment or final self-efficacy score cannot show whether they spent the semester gradually moving from observation to hands-on tool use, whether they relied heavily on help-seeking early and became more independent later, or whether stress was associated with productive persistence or disengagement. Similarly, simply counting how often a student uses a machine or collaborates with peers may miss the order of behaviors. In makerspaces, sequence matters: observing a machine before using it means something different from observing it after a failed attempt. Seeking help before trying a tool may represent preparation, while seeking help after repeated attempts may represent regulation or troubleshooting.

Recent advances in multimodal learning analytics make it possible to study these kinds of patterns at scale. Video-based pose estimation, for example, allows researchers to extract information about students' physical movement and location without relying only on manual observation. Computer vision tools can estimate human body keypoints from video, creating pose data that can be used to study embodied activity in complex environments (Fang et al., 2022). In makerspaces, this is especially promising because learning is deeply physical and spatial. Students move between machines, tables, peers, and materials; they stand, observe, manipulate tools, collaborate, and work alone. These embodied patterns may reveal forms of participation that are difficult to capture through surveys or written reflections alone (Adler-Belendez & al., 2020).

However, pose data are not meaningful on their own. A coordinate, posture, or movement pattern becomes educationally useful when it is interpreted in relation to learning theory and context. For this reason, pose-based analytics should not be treated simply as a way to automate observation. Instead, they should be used to construct theoretically meaningful representations of student activity. In this study, pose data were used to derive behavioral profiles that describe different ways students participated in the makerspace. These profiles allow us to ask whether

students who show different embodied participation patterns also differ in self-efficacy, belonging, stress, enjoyment, and work habits. This approach treats individual differences not as fixed traits, but as patterns of participation that emerge across the semester.

A second methodological challenge is that makerspace behaviors are temporal. Students move among widely different behavioral states: they observe, interact, collaborate, seek help, work independently, return to machines, and shift between central and peripheral participation. Ordered Network Analysis (ONA) is useful for this purpose because it represents ordered relationships among coded events rather than only their frequency (Tan et al., 2022). ONA allows researchers to model how behaviors are connected across time, producing networks in which nodes represent behavioral states and edges represent temporal associations among those states. This makes it possible to compare not only whether students engage in certain behaviors, but also how those behaviors are organized.

For makerspace research, this is important because learning may be reflected in transitions. A student who repeatedly moves from observing a machine to interacting with it may be developing confidence through vicarious and mastery experiences. A student whose machine interaction is frequently connected to help-seeking may be navigating technical uncertainty through social support. A student whose tool use is followed by solitary peripheral work may be regulating, reflecting, or recovering from a frustrating experience. These interpretations require caution, but they demonstrate why temporal behavioral structure may provide insight into self-efficacy development. ONA provides a way to quantify and visualize these structures while keeping the analysis close to the sequence of student activity.

2.4 Summary

Taken together, this work builds on three bodies of literature. First, research on constructionism and makerspace learning shows that making can support powerful forms of learning, identity development, and participation, but that these experiences vary across students. Second, self-efficacy theory provides a framework for understanding why students' confidence, help-seeking, observation, persistence, and affective experiences matter in open-ended learning environments. Third, multimodal and temporal learning analytics provide tools for studying how these processes unfold through embodied behavior over time. The contribution of this study is to bring these perspectives together. We use pose-based behavioral profiles and Ordered Network

Analysis to examine individual differences in makerspace participation and to connect those differences to self-efficacy-theory-aligned behavioral structures.

This study therefore asks whether students differ in their self-efficacy, stress, enjoyment, or belonging, and also how those differences are related to patterns of embodied participation. By examining students' behavioral profiles and the temporal organization of their makerspace activity, we aim to better understand how students experience open-ended making differently and how those differences may relate to the development of confidence in their own ability to solve problems, persist through challenges, and participate as makers.

2.5 Research Questions

Our research questions (RQs) are as follows:

RQ1. What distinct behavioral and affective profiles emerge from students' pose-based makerspace participation across the semester? In other words, can students be grouped based on behavioral and affective indicators related to mastery, vicarious experience, social support, and affective state?

RQ2. How are these student profiles, individually and in combination, associated with self-efficacy and community belonging? Here, we combine the profile-outcome and composite-profile questions by asking whether single pathways or combined favorable profiles better explain differences in self-efficacy and belonging.

RQ3. How does community belonging shape the relationship between student profiles and self-efficacy? This question examines whether belonging moderates the relationship between students' behavioral/affective profiles and their self-efficacy.

RQ4. How do Bandura-aligned behavioral patterns differ across profiles and evolve over the semester? Here we combine the ONA and longitudinal modeling components by examining whether profiles differ in mastery, vicarious experience, social support, and affective-regulatory behavioral patterns, and whether these patterns change over time.

3. Methods

3.1 Participants

Participants were 56 students enrolled in a digital fabrication course at the graduate school of education of a private university in the United States. Data were collected across three course iterations: Spring 2022 (N = 18), Fall 2022 (N = 19), and Spring 2024 (N = 19). Participants

were between 25 and 35 years old and included 8 male and 48 female students. Most participants had no prior experience with digital fabrication and had not previously used a makerspace. All participants provided informed consent and agreed to the collection of video data while working in the makerspace.

3.2 Procedure

Data collection occurred during three offerings of the same semester-long digital fabrication course. In the course, students were introduced to a range of tools and techniques, including laser cutting, 3D printing, vinyl cutting, sewing, electronics, microcontrollers, and sensing technologies. Enrollment was capped at approximately 20 students per course iteration, which is typical for hands-on digital fabrication courses. During the first half of the semester, students completed weekly mini-projects, each designed to introduce a specific tool or technique. The second half of the semester centered on a final project, in which students designed a learning toolkit or environment of their choice using the resources available in the makerspace. Each week, students were required to write a blog post documenting their makerspace work and were encouraged to comment on their classmates' posts. As part of these weekly blog posts, students also reported their level of enjoyment, stress, and time spent working on the course on a scale from 0 to 5.

3.3 Design

The data collected for this study included pre- and post-surveys, as well as periodic surveys administered throughout the semester. Students' self-efficacy was measured using the validated General Self-Efficacy Scale (Schwarzer & Jerusalem, 1995), which includes items relevant to makerspace contexts, such as "I can always manage to solve difficult problems if I try hard" and "When I am confronted with a problem, I can usually find several solutions." Sense of community belonging was measured using the Brief Sense of Community Scale (Peterson et al., 2008), which includes items such as "I feel like a member of [this makerspace]" and "I have a good bond with others in [this makerspace]." In addition, each week students self-reported their levels of enjoyment and stress on a 1–5 Likert scale when submitting their project descriptions, along with the number of hours they spent working on the assignment. For the purposes of this paper, we analyzed students' average enjoyment, stress, and weekly work hours across the semester, as well as self-efficacy and community belonging gains. We used gain scores rather than including pre-test scores as covariates to avoid unnecessary model complexity, given the

relatively small sample size. Using both pre- and post-test scores as covariates can also introduce multicollinearity; gain scores reduce this relationship to a single interpretable measure while avoiding these issues.

3.4 Data Collection, Preprocessing and Analysis

The makerspace was equipped with eight high-resolution cameras that captured synchronized video recordings during its operating hours (Fig. 2). Across each semester, video data were collected over a 14-week period, resulting in approximately 10 TB of high-resolution video files. These recordings were first processed to extract 2D pose data, consisting of the x and y coordinates of 17 body joints, using AlphaPose (Fang et al., 2022). The resulting 2D pose data were then used to reconstruct 3D pose data with Poseconnect, developed by the Wildflower group (Quinn, 2024).

This subsection describes how we constructed student-level behavioral profiles from pose- and gaze-derived makerspace data. The purpose of the profile analysis was to summarize how students combined different forms of participation across the semester, rather than to test each behavioral indicator separately. Guided by Bandura's self-efficacy framework, we mapped behavioral and self-report indicators onto mastery, vicarious, social, and affective pathways, then used a person-centered clustering procedure to identify recurring configurations of these indicators.

3.4.1 Student Profiles

After reconstructing the pose and gaze data, we aggregated students' behavioral traces into semester-level indicators and used them to construct Bandura-informed student profiles. The purpose of this analysis was to identify descriptive profile dimensions from behavioral and affective indicators.

3.4.2 Profile Indicator Aggregation

The pose and gaze processing described above produced per-second behavioral labels for each student. For the profile analysis, these labels were aggregated to the student-semester level. The main pose-based indicators used here captured machine interaction, machine observation, joint visual attention, and peer-proximity behavior. These semester-level indicators were then joined with weekly enjoyment and stress self-reports.

These indicators were treated as behavioral proxies for Bandura's self-efficacy pathways, not as direct measures of students' internal experiences. Machine-use seconds do not distinguish fluent tool use from struggle, shared gaze does not verify joint task focus, and machine observation does not capture all forms of vicarious learning. The count-based indicators were raw second-level totals rather than rates normalized by time present. The log transformation described below reduces skew in these totals, but it does not remove all exposure-related differences.

3.4.3 Bandura-Aligned Feature Construction

We next translated the student-semester indicators into Bandura-aligned candidate features. Bandura (1997) describes four pathways through which people build self-efficacy: mastery experience, vicarious experience, social persuasion, and affective state. For the profile analysis, each engineered feature was mapped to one of these pathways and standardized before clustering.

Indicators were first transformed using $\log(1 + x)$ to reduce skew, and then standardized to a mean of 0 and a standard deviation of 1. Weekly enjoyment and stress ratings were standardized directly. The resulting candidate features included a mastery composite, a machine-observation feature, joint visual attention, a social composite, stress, and an affective composite used for outlier screening**. Table 1** gives the definitions used in the analysis.

Redundant and tool-idiosyncratic inputs were excluded before clustering. For example, collaboration was excluded because it duplicated joint visual attention, and laser-cutter observation was excluded because it often reflected passive waiting rather than active observation.

Table 1

Feature engineering: mapping tracked behavior to Bandura's four self-efficacy pathways

Bandura pathway	Engineered variable	Definition	Note

Mastery	M_comp	mean of z_doing and z(enjoyment), where z_doing is the mean of z(log(1 + machine interaction seconds)) and z(log(1 + individual machine-use seconds))	Weekly enjoyment serves as the success signal
Vicarious	z_observe	z(log(1 + seconds observing the tool stations))	Laser-cutter observation excluded as passive waiting
Social	z_jva	z(log(1 + joint visual attention seconds))	Carried forward from prior work.
	z_social	mean of z_jva and z(log(1 + cooperation seconds))	Collaboration excluded as redundant with joint visual attention (r = .983)
Affective	z_stress	z(weekly stress rating)	—
	z_affect	z((z(enjoyment) - z(stress)) / 2)	Used only for outlier screening

Note. Candidate engineered variables and their construction are shown above; per-feature screening statistics and the full tracked-column list appear in the supplementary materials, and retained combinations are reported in Table 4. z denotes standardization to a mean of 0 and a standard deviation of 1; log(1 + x) is applied to skewed second-level counts before standardizing. Enjoyment and stress are weekly self-reports (ranging from 1 to 5).

3.4.4 Profile Dimension Derivation

We used K-means clustering (MacQueen, 1967) for building student profiles, and treated each grouping as a descriptive partition evaluated through resampling stability rather than as an inferred latent class. Candidate feature sets were required to span at least two Bandura pathways; the search over them was exploratory, and its association p-values were used only to rank candidates. For each candidate set we evaluated K from 2 to 5 and kept the solution with the

highest silhouette among those whose smallest group held at least 10 students. Clustering used a fixed seed and twenty initializations, held both outcomes out of the feature space, and ordered labels by centroid so that the favorable side of each dimension is fixed from theory.

Each candidate solution was then evaluated on three diagnostics: the silhouette score for separation (Rousseeuw, 1987), the mean bootstrap Adjusted Rand Index against the full-sample solution for stability (Hubert & Arabie, 1985; Hennig, 2007), and a cluster-by-cohort chi-square for balance. Because the three cohorts differed in instructors, projects, and room layout, features whose variance was largely explained by cohort (eta-squared of approximately .30 or above) were screened out, so that semester differences could not masquerade as individual differences; the clearest case was a peer-observation count that primarily indexed semester membership (cohort eta-squared = .89). **Table 2** summarizes these modeling decisions and their purposes.

Table 2

Modeling decisions in the profile derivation and screening procedure

Modeling decision	**Why it was used**
Candidate sets had to span at least two Bandura pathways	The goal was to build student profiles from combinations of theoretically meaningful indicators, not to split students by one isolated behavior.
Clustering algorithm, (K-means, K = 2 to 5, n_init = 20, fixed seed)	Built descriptive student profile dimensions
K was selected using silhouette with a minimum group size of 10	Balanced separation against interpretable group size
Candidate solutions were evaluated for stability and cohort balance. Mean bootstrap Adjusted Rand Index across 100 subsamples (80% without replacement)	Assessed whether partitions were stable

Features with high cohort dependence (cohort eta-squared approximately .30 or above) were flagged or excluded; borderline cases documented in supplement	Reduced the risk of recovering semester differences rather than individual-difference profiles. The three cohorts differed in instructors, projects, and room layout, so features that mainly indexed semester membership were not suitable anchors.

Note: These decision rules were applied before interpreting any profile-outcome associations. Borderline cases and full screening statistics are reported in the supplementary materials.

The Social/JVA dimension was handled separately. Prior work on this dataset theorized that joint visual attention operates on self-efficacy through community belonging. We therefore retained JVA as the social dimension in the profile structure. Applying K-means with K = 2 to JVA reproduced the same High/Low JVA grouping, so this dimension was included in the primary three-dimension composite reported below. The retained dimensions, their profiles, and their diagnostic values are reported in the Results. After profiles were formed, dimension-level differences on each outcome were tested with two-group Welch t tests and reported as Cohen's d with 95% confidence intervals.

3.4.5 Composite Profile Construction

Bandura (1997) describes the four pathways as operating together within a person rather than in isolation, so beyond testing each dimension separately we also tested whether holding favorable profiles on several dimensions at once was associated with self-efficacy. A composite profile is the count of a student's favorable profiles. Here, "favorable" refers to the side of a dimension that theory links to higher self-efficacy. For example, a student who is favorable on one dimension and unfavorable on the other two receives a composite score of 1, while a student who is favorable on all three dimensions receives a score of 3. Because the count is derived from fixed profile labels, it does not introduce additional fitted parameters. Its interpretation, however, remains tied to the exploratory profile-selection process.

We formed a three-dimension composite as the primary composite. It counted favorable profiles across the Affective, Behavioral, and Social dimensions, producing a score from 0 to 3.

This composite was used as the main profile measure because it represented the full Bandura-aligned profile structure in the study. Because a composite is an ordered count, its association with each outcome was assessed as a linear trend across the ordered levels, with the omnibus analysis of variance reported alongside. Both composites were tested against both outcomes, so that the composite and belonging are empirically separable before they enter one model.

3.4.6 Derivation of Ordered Network Analysis Features

Using the previously collected makerspace behavioral trace dataset and previously derived student profile labels, we constructed Ordered Network Analysis features to represent the temporal organization of student behavior. ONA was used to model how behavioral states and machine-related actions were sequentially associated within student sessions (Tan et al., 2023). ONA has the benefits of accounting for the temporal nature between connections, while also being highly visually interpretable. Networks were constructed in R using the tma and ona packages and all data preprocessing was done using the tidyverse suite of packages.

Each timestamped row was converted into a collection of binary behavioral codes corresponding to the state the student is in at that moment. Categorical states were one-hot encoded, and machine related behaviors were condensed into generic observe- and interact-machine indicators. Our unit of analysis is segmented at the student and session level, where each unit corresponds to a unique combination of student and session identifier. ONA then uses a moving window to establish connections from the current timestamp to the previous timestamps in the dataset. We used a window size of seven observations, consistent with prior ONA applications and selected to capture temporally proximal behavioral associations while limiting the influence of distant events.

Whereas the behavioral profiles were derived from semester-level summaries of full-rate behavioral duration, ONA characterized the temporal organization of behavioral transitions. To prevent long strands of repeated behaviors from dominating the network, we downsampled the behavioral stream to retain only rows where the observed behavioral state changed within each observation unit. This emphasized transitions among meaningful behavioral states rather than duration within a repeated state. For each student-session unit, this procedure produced a weighted network in which nodes represented behavioral codes and edges represented

temporally accumulated associations among codes. These edge weights were then used as quantitative features in subsequent between-profile comparisons and longitudinal regression analyses.

3.4.7 Derivation of Pillar Scores

Since Bandura's self-efficacy theory combines multiple behavioral tendencies under four distinct pillars (Bandura, 1977), we aggregated conceptually related pair-wise ONA connections into pillar scores relating to mastery, vicarious experiences, affective experiences, and social support/persuasion experiences. The reason why we do this is twofold: individual edge weights can be sensitive to sparsity and session-specific behavioral variation, and self-efficacy can manifest in makerspace environments in different ways. This approach allows us to conduct analyses between groups on higher level constructs rather than many individual edges. Therefore, we assign these transitions with rationale in the tables below.

Table 3

Summary of ONA edge mappings to Bandura-aligned pillars

Pillar score	Included edge patterns	Rationale
Vicarious experience	Repeated machine observation; machine observation connected with collaboration or cooperation	Observation occurring independently or within socially coordinated activity
Mastery experience	Repeated machine interaction; machine observation and interaction in either order	Sustained tool use and movement between observing and hands-on activity

Social support and persuasion proxy	Help-seeking connected with machine observation, machine interaction, collaboration, or cooperation	Social support occurring around tool use and peer activity
Affective and regulatory proxy	Machine interaction connected with solo peripheral activity or help-seeking	Possible regulation, uncertainty, or movement into and out of active tool use

For each student-session unit, we first obtained the ONA edge weights representing the temporally accumulated weight of observed behavioral transitions. We then used the table above to assign selected transitions to Bandura-aligned pillars. Each pillar score was computed as the summed weight of the transitions mapped to that pillar, producing a lower-dimensional construct-level summary of the unit's temporal behavioral organization. Formally speaking, a pillar score can be computed as

$$P_{u,k} = \sum_{e \in E_k} w_{u,e}$$

where E_k is the set of theory-selected edges mapped to pillar k, and $w_{u,e}$ is the ONA edge weight for edge e in student-session unit u.

3.4.8 Profile-level Comparisons of Pillar Scores

In order to examine whether the derived student behavior profiles differed in their self-efficacy-theory-aligned ONA structure, we compared pillar scores across the behavioral and affective profile categories: deep engagers versus surface engagers, and stress-burdened versus stress-regulated students. For each Bandura-aligned pillar, we conducted group comparisons using

mean pillar scores. We used Welch independent-samples t-tests for two-profile contrasts and one-way ANOVA models when comparing across multiple categories. These analyses provide an initial semester-level assessment of whether the profiles differed in the strength of theoretically relevant state transitions.

3.4.9 Longitudinal Modeling of Pillar Scores

While the average profile-level comparisons give us insight into how the student profiles differ at the semester level, we are also interested in understanding how these groups evolve behaviorally over the course of the semester. To accomplish this, we aggregated the pillar scores by week and utilized a generalized additive mixed model (GAMM), fit in R using the mgcv package, for regression analysis and inference. GAMMs extend generalized linear modeling by allowing predictors, such as time, to enter the model through smooth functions rather than only through fixed linear effects. This allows us to model the nonlinear behavioral trajectories throughout the duration of the semester.

Specifically, we modeled weekly log-transformed pillar scores as a function of profile membership, semester, and profile-specific smooths over normalized semester time. This allowed us to not only examine the average differences between the profiles, but also whether the shapes of their behavioral trajectories differed over the course of the semester. Additionally, to account for repeated measures of the same student, we included student-level random effects. Formally, our model took the form:

log1p(pillar score) ~ profile + semester + s(week_prop, by = profile, k = 5) + s(person_identity, bs = "re")

Separate models were fit for the behavioral and affective profile labels. The term s(week_prop, by = profile, k = 5) specifies separate smooth functions of normalized semester time for each profile group, with the basis dimension set to k=5. This allows the shape of the behavioral trajectory to vary by profile using relatively simple nonlinear forms. The term s(person_identity, bs = "re") specifies a student-level random effect, accounting for repeated weekly observations from the same student.

4. Results

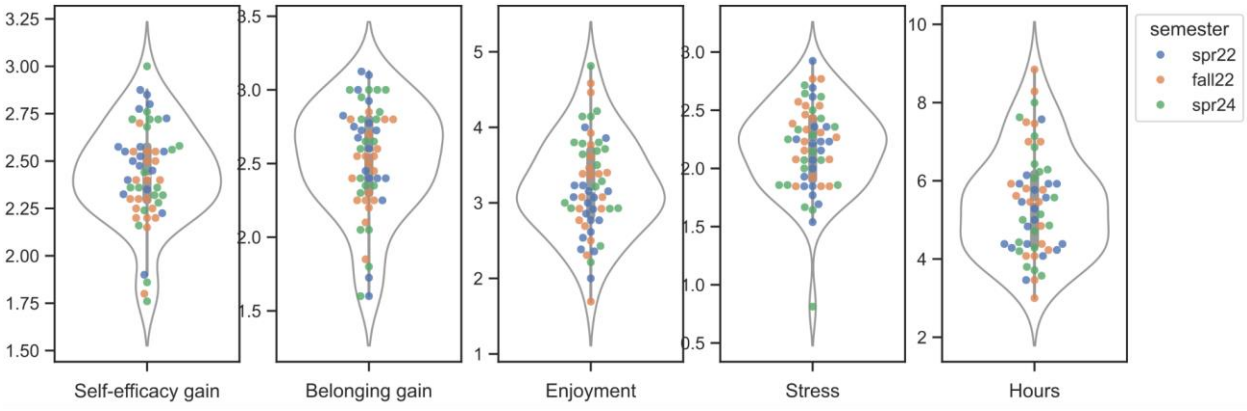
4.1 Descriptive Statistics

Before analyzing the data, we conducted an exploratory data analysis. We first assessed the distributions of our main outcome measures and tested their normality. Figure 1 presents the distributions of each dependent measure; visual inspection indicated no ceiling or floor effects. We then formally tested the normality of each variable using the normaltest function from the SciPy Python library, which combines skewness and kurtosis into an omnibus test of normality (D'Agostino & Pearson, 1973). None of the tests reached statistical significance, suggesting that all five variables were approximately normally distributed.

We also examined whether there were statistically significant differences across semesters using analyses of variance (ANOVAs). No significant differences were found across semesters, with the exception of weekly reports of enjoyment, $F(2, 53) = 3.45$, $p < .05$, $\eta^2 = .11$. Post hoc tests indicated that enjoyment ratings collected in Spring 2024 ($M = 3.38$, $SD = 0.65$) were significantly higher than those collected in Spring 2022 ($M = 2.94$, $SD = 0.49$). Overall, these results suggest that the data collected across the three semesters were largely comparable. Therefore, we combined the three semesters into a single dataset for the analyses reported below.

Figure 1

Distribution of the main variables.



Note. Self-efficacy and belonging indicate normalized scores at the end of the semester; enjoyment, stress, and hours spent are weekly averages. We show individual data points (colored by semesters) to show that the three cycles of data collection are comparable.

4.2 Student Profiles Results

4.2.1 Student profiles reveal an affective and a behavioral dimension, each related to self-efficacy

The clustering procedure for the student profile construction retained two clustered dimensions and one pre-specified Social dimension. The first clustered dimension separated students by behavioral engagement. One group was above the sample average on both tool observation and hands-on mastery activity, with centroids of +0.92 and +0.41 standard deviations. The other group was below average on both indicators. Because observation and hands-on activity rose together in the first group, we labeled the two profiles Deep Engagers and Surface Engagers. This dimension captured a pattern of deeper engagement with tools, rather than observation or hands-on activity alone.

The second clustered dimension separated students by affective regulation. One group reported lower stress, with a centroid of -0.62, while the other group reported higher stress, with a centroid of +0.78. The lower-stress group also showed slightly higher social contact and slightly lower mastery activity, but the split was mainly driven by stress. We therefore labeled the two profiles Stress-Regulated and Stress-Burdened. The Affective dimension was less cleanly separated than the Behavioral dimension, but it remained stable under resampling.

The Social dimension split students into High and Low joint visual attention. This dimension was carried forward from prior work rather than selected through the exploratory clustering search. Together, the Behavioral, Affective, and Social dimensions formed the profile structure used in the composite analyses below. These dimensions should not be read as pure one-to-one measures of Bandura's pathways: the Behavioral dimension combined mastery and vicarious indicators, the Affective dimension was stress-dominant but also included mastery and social inputs, and the Social dimension used joint visual attention.

Table 4

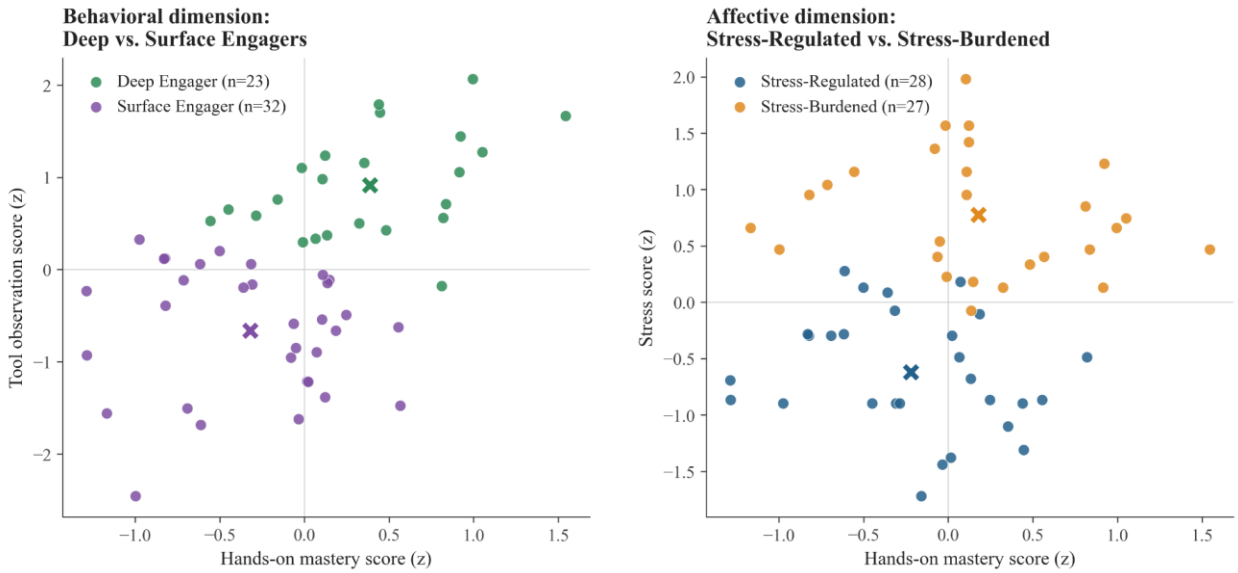
Dimensions: clustering inputs, quality, and single-dimension test against self-efficacy

Dimension	Clustering inputs (pathways combined)	Profiles (n)	Silhouette	Stability (ARI)	Cohort balance (χ^2 p)	Single dimension to SE, d [p]
Affective	z_stress, M_comp, z_social (Affective, Mastery, Social)	Stress-Regulated (28) / Stress-Burdened (27)	.28	.78	.114	+0.56 [.043]
Behavioral	z_observe, M_comp (Vicarious, Mastery)	Deep Engager (23) / Surface Engager (32)	.42	.91	.030	+0.51 [.057]
Social	z_jva (Social)	High JVA (22) / Low JVA (33)	—	—	—	−0.03 [.921]

Note. N = 55. The Affective and Behavioral dimensions are K-means groupings; the Social dimension is a pre-specified single-feature split in joint visual attention. Silhouette ranges from -1 to 1, with higher values indicating cleaner separation; ARI is the mean Adjusted Rand Index across 100 resamples at 80% without replacement, with higher values indicating greater stability. Given its lower silhouette, the Affective grouping is read as exploratory. Sensitivity: the affective outlier (z = 4.31) does not need to be excluded for the non-affective Behavioral dimension; retaining that participant gives Deep Engager 24 / Surface Engager 32, ARI = .90, cohort chi-square p = .06, and d = +0.58, 95% CI [+0.04, +1.12], p = .032. These single-dimension tests came from an exploratory search and are descriptive.

Figure 2

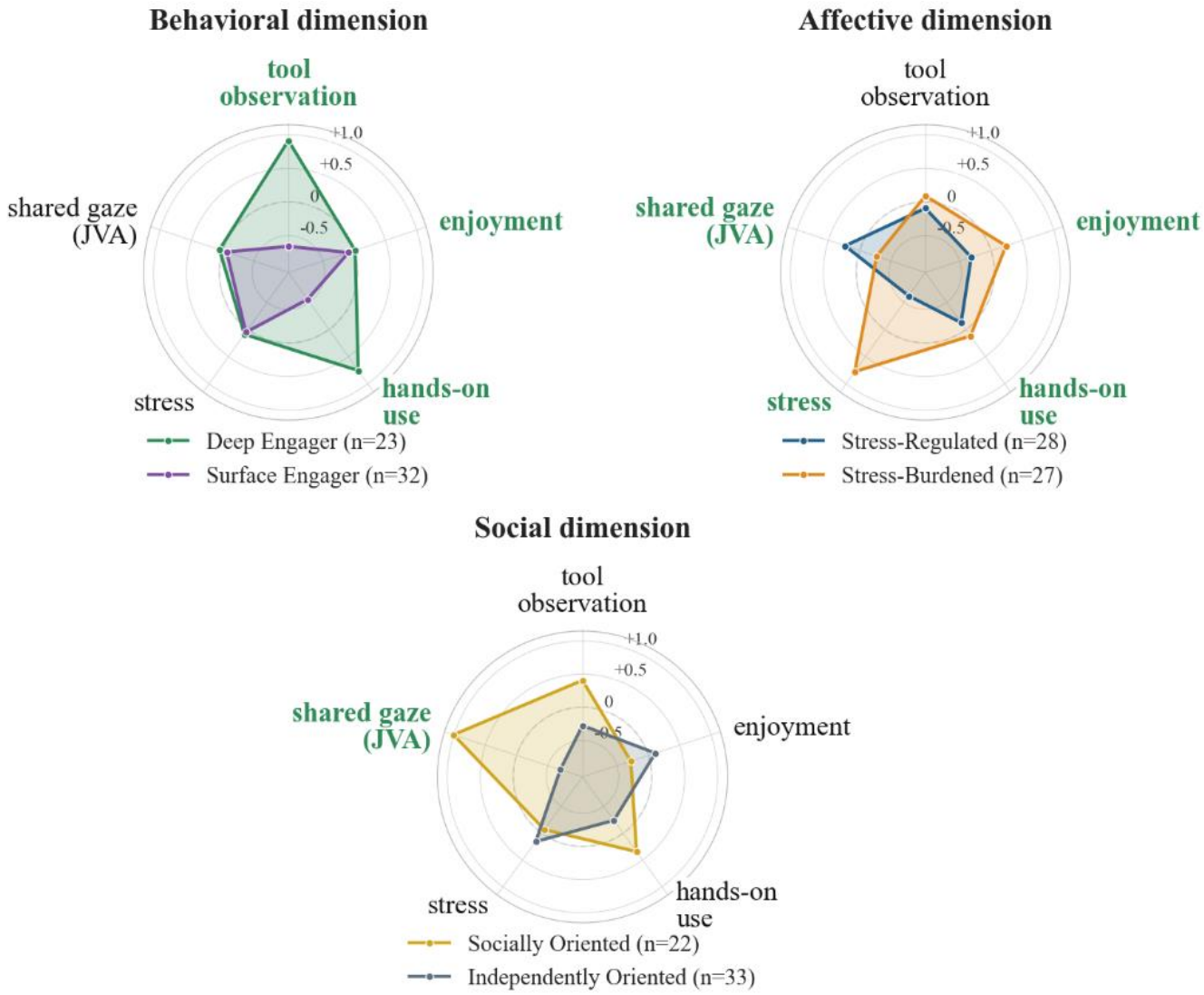
Scatter plots of students' profiles based on their mastery / observation scores.



Note: Each dot is one student. Left: the behavioral dimension, with Deep Engagers higher on both hands-on machine use and tool observation than Surface Engagers. Right: the affective dimension, with Stress-Regulated students lower on stress than Stress-Burdened students; the two groups overlap on engagement. The X marks each group mean. Scores are standardized (0 = sample mean).

Figure 3

Radar chart of the three student profile dimensions



When tested one dimension at a time, the clearest self-efficacy difference appeared in the Affective dimension. Stress-Regulated students reported higher self-efficacy than Stress-Burdened students, d = +0.56, 95% CI [+0.02, +1.10], p = .043. Deep Engagers also reported higher self-efficacy than Surface Engagers, but this contrast did not reach conventional significance in the main analytic sample, d = +0.51, 95% CI [-0.04, +1.05], p = .057. The sensitivity check retaining the affective outlier showed the same directional pattern with a stronger estimate, d = +0.58, p = .032. The Social dimension showed no direct self-efficacy difference, d = -0.03, 95% CI [-0.57, +0.51], p = .921.

We did not observe corresponding associations with community belonging, Affective p = .71, Behavioral p = .32, Social p = .35. The profile dimensions therefore showed clearer links to self-efficacy than to belonging, leaving belonging to be examined separately in the regression model below.

4.2.2 Combining profile pathways relates to self-efficacy

Because the single-pathway profiles gave only partial signals, we next tested whether combining profile labels provided a clearer pattern. We first counted how many of the two clustered pathways, the affective and the behavioral, fell on each student's favorable side, and then extended the count to all three dimensions. Across the three profiles (Stress-Regulated, Deep Engager, and High joint visual attention), 11 students held no favorable profile, 21 held one, 17 held two, and 6 held all three.

Students with more favorable profiles reported higher self-efficacy, although the association was modest. Across the ordered composite levels, self-efficacy showed a positive linear trend, r = .28, 95% CI [.01, .51], p = .04. Most of the increase occurred between zero and one favorable profile: mean self-efficacy rose from 2.24 among students with no favorable profile to 2.46 among students with one favorable profile, then remained similar among students with two profiles, M = 2.46, and all three profiles, M = 2.49. Because only 6 students held all three favorable profiles, this leveling should be read as a descriptive pattern rather than evidence of a tested threshold.

The omnibus test across the four levels did not reach conventional significance, but the planned linear trend did, which matches the observed zero-to-one increase and later leveling (see the Table 5 note). Adding the Social dimension did not increase the raw association, which is what would be expected if the social pathway acts on self-efficacy through belonging rather than directly; its value surfaces once belonging is modeled. The primary three-profile composite showed a modest association, interpreted through its effect size and confidence interval rather than the .05 threshold alone. The two-profile count over the two clustered dimensions alone shows the same pattern while the gradient, however, contained consistent exceptions, which we examine next.

The two-pathway composite was associated with self-efficacy, $F(2, 52) = 6.90$, $p = .002$, $r = .41$. This was stronger than the strongest single-profile result. Most of the gain appeared between zero and one favorable profile, with less additional gain from holding both. The composite was not associated with community belonging, $r = .15$, $p = .28$. We then added the already validated social pathway (JVA) to complete the three-pathway student profile.

Table 5

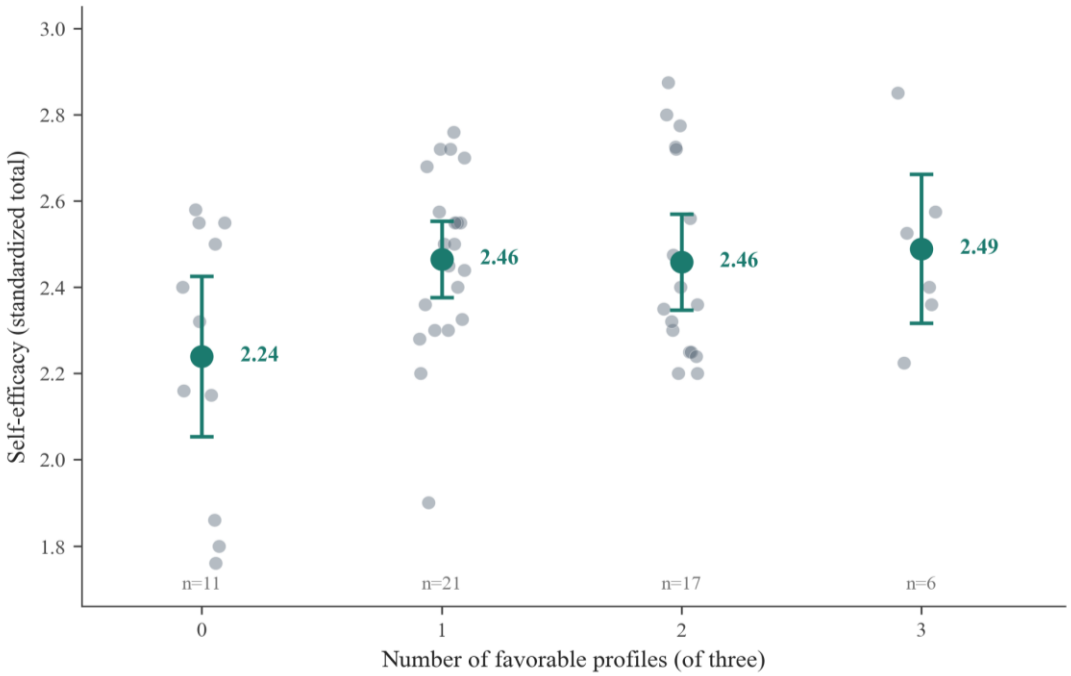
Composite scores and their relationship to self-efficacy and belonging.

Composite to outcome	F (df)	p	η^2	ω^2	r [95% CI]
Three-dimension to self-efficacy	2.61 (3, 51)	.061	.13	.08	.28 [.01, .51]
Three-dimension to belonging	0.66 (3, 51)	.580	.04	−.02	.17 [−.10, .42]

Note. N = 55. The three-dimension composite (Affective + Behavioral + Social) is the primary composite. Its linear association with self-efficacy is significant ($r = .28$, $p = .04$); most of the rise occurs between zero and one favorable dimension. The composite is not associated with belonging, so it indexes self-efficacy specifically. ω^2 corrects the small-sample upward bias of η^2.

Figure 4

Self-efficacy by number of favorable profiles.



Teal points are group means with 95% confidence intervals; grey dots are individual students.

4.2.3 Belonging Helped Explain Exceptions in the Three-dimension Profile Composite

Across the three-profile composite, self-efficacy followed a general gradient, but several exceptions pointed to the same factor. Among the 11 students with no favorable profile, 4 still reported relatively high self-efficacy, at 2.50 or above. Each of these students also reported belonging at or above the sample median. At the other end of the gradient, among the 6 students favorable on all three profiles, the student with the lowest self-efficacy also reported the lowest belonging in the sample, tied with one other student.

These exceptions were small in number and should be read descriptively. However, they all pointed in the same direction. Departures from the profile gradient consistently involved students' sense of belonging. This pattern suggested that whether the composite tracked self-efficacy depended not only on profile count, but also on belonging. We therefore examined belonging directly in the regression analysis that follows.

4.3 Multi-Regression Model with Community Belonging

To test whether community belonging helped account for differences in self-efficacy beyond the profile composite, we regressed self-efficacy on a composite, belonging, and their product, with all predictors mean-centered (Aiken & West, 1991):

$$SE = b_0 + b_1 C_c + b_2 BEL_c + b_3 (C_c \times BEL_c) + e$$

All predictors were mean-centered before the product term was computed. This kept the main coefficients interpretable at the average level of the other predictor and reduced nonessential collinearity with the interaction term. Variance inflation factors are reported with the moderation regression table.

The belonging coefficient estimates the association between belonging and self-efficacy after accounting for the profile composite. The product term tests whether the association between the composite and self-efficacy differs depending on students' level of belonging. Simple slopes were computed at one standard deviation below and above the belonging mean, with a median-split correlation used as a descriptive check. Additionally, residuals were checked for normality using the Shapiro-Wilk test and for constant variance using the Breusch-Pagan test. The simple count model with continuous belonging was also compared against weighted-profile, cluster-distance, and belonging-binarized alternatives; those comparisons are reported in the supplementary materials.

4.3.1 Profiles mattered for self-efficacy among students with lower belonging

The composite was most strongly related to self-efficacy among students with lower community belonging. We added community belonging to the composite model as a covariate, along with the composite-by-belonging product term. Adding belonging increased the variance explained in self-efficacy from $R^2 = .08$ to $R^2 = .32$, and adding the product term increased it further to $R^2 = .38$. The product term was significant, b = -0.151, 95% CI [-0.283, -0.018], p = .027.

The direction of the product term indicated that the composite mattered more when belonging was lower. One standard deviation below the belonging mean, each additional favorable profile was associated with about +0.10 in self-efficacy, p = .009. One standard deviation above the belonging mean, the slope was approximately zero, p = .81. A median split on belonging showed the same pattern descriptively: below the belonging median, the composite

accounted for about 14 percent of the variance in self-efficacy, whereas above the median it accounted for almost none.

The same pattern was visible in the group means. Among lower-belonging students, those with no favorable profile averaged 2.12 in self-efficacy, while those with at least one favorable profile averaged 2.40, a gap of +0.28. Among higher-belonging students, those with no favorable profile averaged 2.54 and those with at least one favorable profile averaged 2.52, a gap of -0.02. Thus, when belonging was low, favorable profiles distinguished students with higher and lower self-efficacy; when belonging was high, self-efficacy remained similar regardless of profile count.

This pattern is consistent with the interpretation of community belonging as a protective layer in the profile-self-efficacy relationship. The two-profile sensitivity composite showed the same product-term pattern, b = -0.183, 95% CI [-0.360, -0.006], p = .043. Full regression estimates and sensitivity checks are reported in the regression and supplementary sensitivity tables.

Table 6

Regression of self-efficacy on the three-dimension composite, belonging, and their interaction (mean-centered predictors)

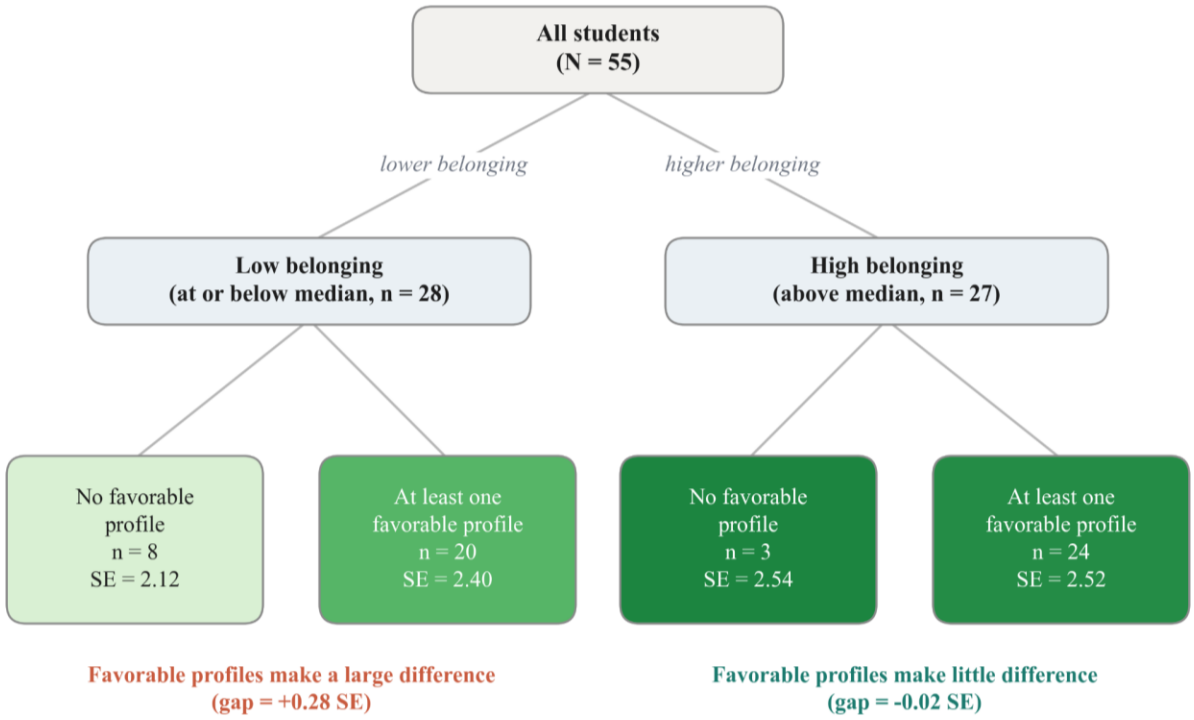
Term	Additive	Interaction
Composite b [95% CI]	+0.052 [−.012, .116]	+0.046 [−.015, .108]
Belonging b [95% CI]	+0.335 [.178, .493]***	+0.362 [.209, .515]***
Composite × Belonging b [95% CI]	—	−0.151 [−.283, −.018]*
R^2 (adjusted)	.32 (.29)	.38 (.34)

Composite slope, low belonging (−1 SD)	—	+0.10
Composite slope, high belonging (+1 SD)	—	−0.01

Note. N = 55. * p < .05, *** p < .001. Composite-only baseline R^2 = .08. Belonging and the interaction are significant; the composite slope is positive at low belonging and near zero at high belonging, so the composite's association with self-efficacy is conditional on belonging. Residual checks passed (Shapiro-Wilk p = .89; Breusch-Pagan p = .73).

Figure 5

Belonging decides whether the profile matters for self-efficacy



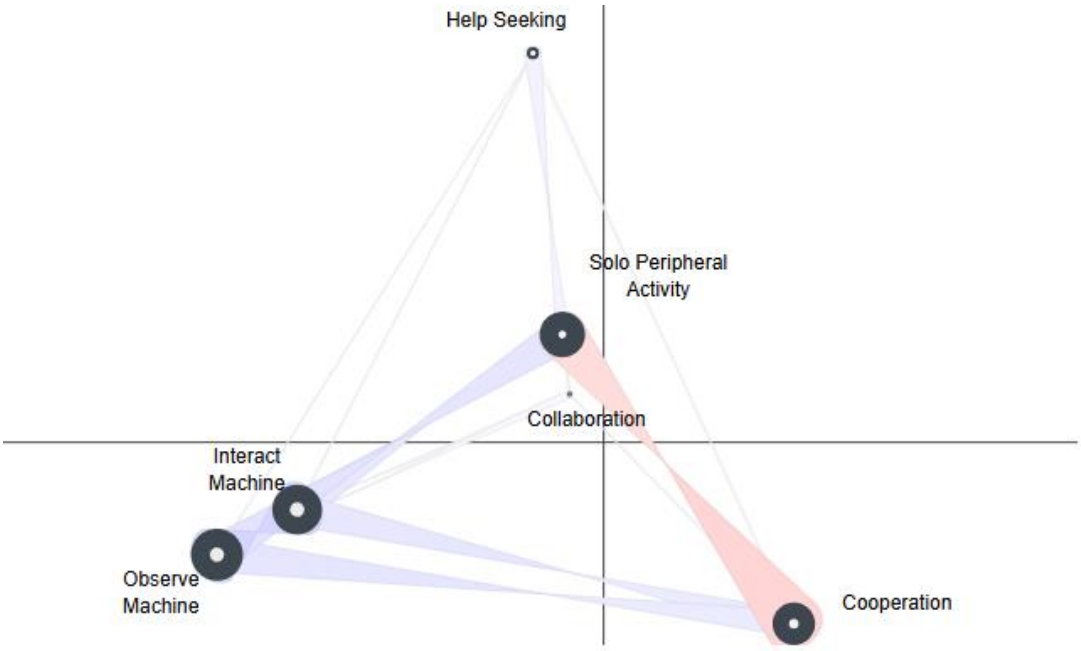
Students split first on belonging, then on whether they hold any favorable profile. Higher belonging holds self-efficacy up regardless of profile.

4.4 ONA Composite Score Results

After identifying which profiles were related to self-efficacy, we used ONA to examine whether these profile labels reflected different temporal behavior patterns in the makerspace. Preliminary comparisons between the Bandura pillar scores indicated no significant differences between the stress-burdened and stress-regulated profiles across the four composite scores, ps = 0.37-0.77. Therefore, subsequent analyses focus on differences between the Deep Engager and Surface Engager behavioral profiles.

Figure 6

Mean-subtracted network between deep and surface engagers.



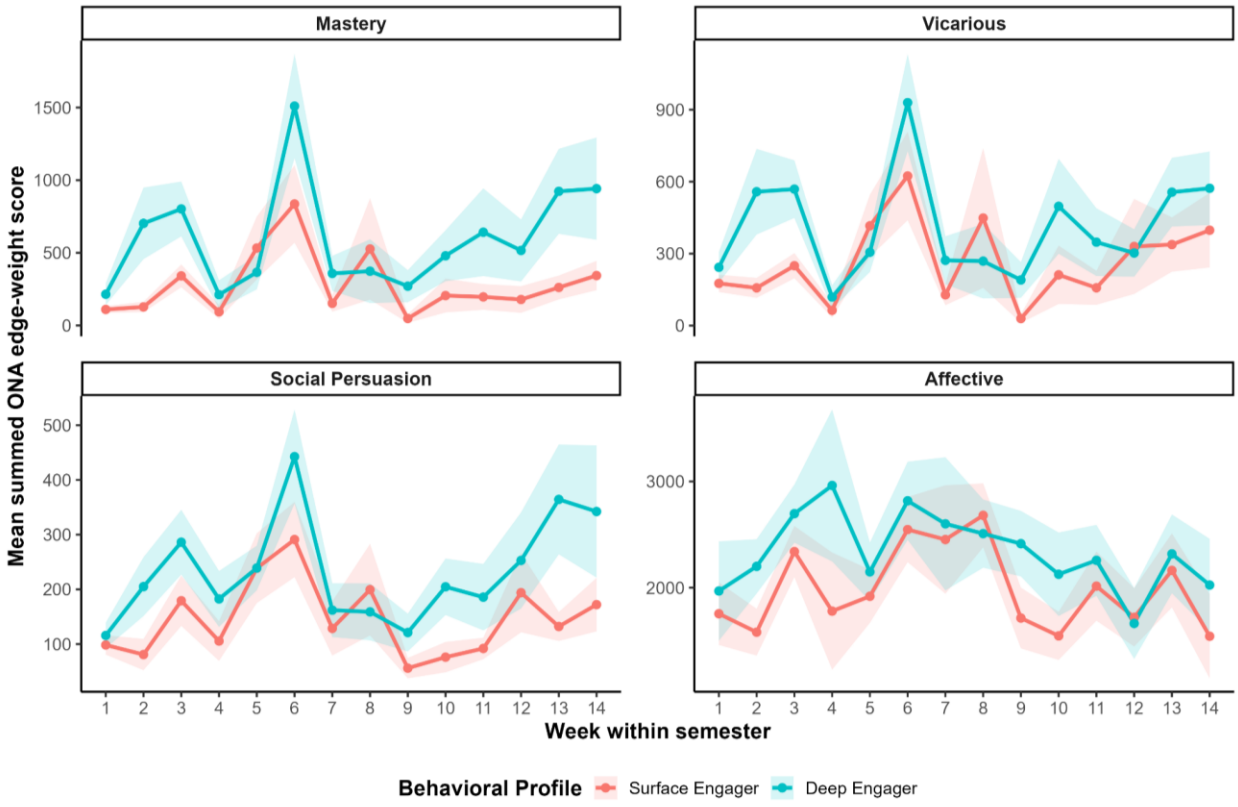
The figure above presents a mean-subtracted ONA network comparing these two profiles, with blue edges indicating stronger connections for Deep Engagers and red edges indicating stronger connections for Surface Engagers. Visual inspection of the mean-subtracted network indicate that deep-engagers exhibited stronger connections among behaviors that support mastery (M_{Deep} = 645.54, $M_{Surface}$ = 297.36, $p < 0.05$), social persuasion (M_{Deep} = 240.62, $M_{Surface}$ = 148.14, $p < 0.05$) and vicarious behaviors (M_{Deep} = 442.26, $M_{Surface}$ = 280.00, $p < 0.05$). Surface Engagers had numerically higher affective-aligned scores (M_{Deep} = 1955.29, $M_{Surface}$ = 2270.09, $p > 0.05$), this difference was not statistically significant.

4.4.1 GAMM Modeling Results

The preceding analyses describe aggregate differences in Bandura composite scores between the behavioral profiles. However, these comparisons do not explain how these scores changed over the duration of the course. To examine the temporal evolution of these patterns, we first plotted weekly mean summed composite scores and then modeled profile-specific trajectories using generalized additive mixed models.

Figure 7

Pooled weekly Bandura composite score trajectories



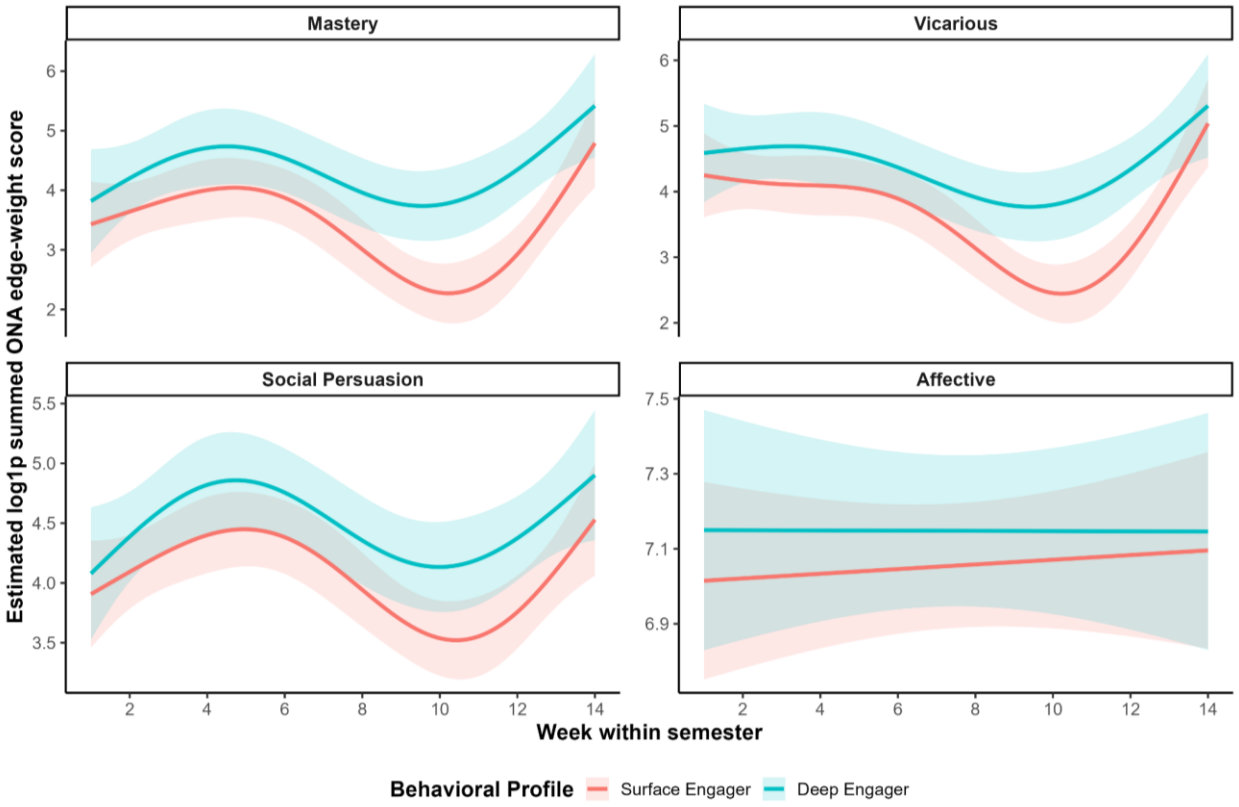
Note: ribbons represent +/- 1 SE. Author, affiliation, email

Temporally, Deep and Surface Engagers appear to follow broadly similar semester-level rhythms, rather than showing a clean divergent growth pattern. However, during the major project period around weeks 5–8, Deep Engagers exhibit higher summed edge-weight activity in the mastery, vicarious, and social persuasion composites compared with Surface Engagers. To test these temporal patterns, we fit generalized additive mixed models predicting weekly log-

transformed Bandura composite scores from behavioral profile, semester, and profile-specific smooths of semester progress. Student identity was included as a random effect to account for repeated observations within students.

Figure 8

Pooled GAMM-estimated Bandura composite trajectories.



Note: ribbons show 95% CIs.

The GAMM-estimated trajectories reinforced the aggregated ONA findings. Across semesters, deep engagers showed consistently higher estimated mastery, vicarious learning, and social persuasion scores than surface engagers, while the affective composite showed minimal separation between profiles. The smooth trajectories also indicated that mastery, vicarious learning, and social persuasion followed nonlinear temporal patterns, with both profiles showing changes around the middle and later portions of the semester. In contrast, affective-aligned scores remained comparatively flat over semester progress.

Table 7

Results of the Bandura pillars based on students' profiles over time.

Bandura pillar	Deep Engager profile effect	Surface Engager smooth over time	Deep Engager smooth over time	Interpretation
Mastery	β = 0.92, SE = 0.23, p < .001	EDF = 3.67, F = 9.24, p < .001	EDF = 3.26, F = 3.76, p = .008	Deep Engagers had higher mastery-aligned scores, and both profiles showed nonlinear temporal trajectories.
Vicarious learning	β = 0.75, SE = 0.21, p < .001	EDF = 3.71, F = 12.58, p < .001	EDF = 3.08, F = 3.77, p = .007	Deep Engagers had higher vicarious-learning scores, with nonlinear trajectories for both profiles.
Social persuasion	β = 0.43, SE = 0.15, p = .005	EDF = 3.54, F = 5.29, p < .001	EDF = 3.29, F = 3.37, p = .015	Deep Engagers had higher social-persuasion scores, and both profiles varied over time.
Affective	β = 0.09, SE = 0.13, p = .494	EDF = 1.00, F = 0.15, p = .697	EDF = 1.00, $F \approx 0.00$, p = .994	No reliable profile difference or meaningful temporal trajectory was detected.

The GAMM trajectories indicate that there is not much separation in the trajectories between deep and surface engagers, only that deep engagers exhibited greater activations of mastery, vicarious, and social persuasion aligned behavioral transitions. In order to validate this, we compared the common smooth against the profile-specific smooths to see accounting for separate smooth trajectories helped in model fit.

Model comparisons did not indicate that profile-specific temporal smooths improved fit over a common temporal smooth for mastery, vicarious learning, and social persuasion, all ps >

0.05. Thus, although the composite scores varied nonlinearly during the semester, there was no evidence that deep and surface engagers' behavioral trajectories diverged during the course. Instead, the GAMM results suggest that the profiles differed primarily in overall level, with deep engagers maintaining higher mastery-, vicarious learning-, and social persuasion- aligned activity over the course.

5. Discussion

This study examined whether pose-based behavioral data can make visible meaningful individual differences in makerspace learning, and whether those differences are related to students' self-efficacy. Across three offerings of a semester-long digital fabrication course, the results revealed distinct behavioral, affective, and social dimensions of participation. Students differed not only in how much they interacted with and observed tools, but also in how stressful they experienced the course and how socially connected their participation appeared to be. Together, these findings suggest that individual differences in makerspaces are better understood as configurations of behavior, affect, and social participation than as differences in activity level alone.

The Affective dimension showed the clearest individual association with self-efficacy. Stress-Burdened students reported lower self-efficacy than Stress-Regulated students, even though the two groups overlapped substantially in their behavioral engagement. This finding is important because it shows that visible participation does not necessarily indicate that students experience their work as successful or confidence-building. A student may continue using tools, completing projects, and interacting with others while also experiencing persistent stress or uncertainty. Instructors who focus primarily on observable productivity may therefore overlook students who remain engaged but feel increasingly burdened by the experience.

The Behavioral dimension provided a complementary perspective. Deep Engagers combined greater hands-on machine interaction with greater observation, rather than replacing observation with action. The ONA results further showed that their participation involved stronger connections among machine observation, machine use, collaboration, and help-seeking. Deep engagement therefore appears to involve more than frequent tool use. It reflects a broader pattern in which students move among observing, acting, and drawing on interpersonal resources.

This result also complicates interpretations of observation as passive or peripheral participation. For Deep Engagers, observation was connected to subsequent tool use and socially coordinated activity. Watching a peer or a machine may therefore be part of an active process of preparing, checking, learning, or deciding what to do next. Similarly, help-seeking and collaboration should not automatically be interpreted as dependence or lack of competence. Their stronger integration into the Deep Engager networks suggests that students may use other people strategically to sustain progress through difficult tasks. Productive makerspace participation may therefore involve knowing when to act independently and when to observe, ask questions, or seek support.

The longitudinal analyses add an important qualification. Deep and Surface Engagers did not follow fundamentally different trajectories over the semester. Both groups responded to similar rhythms in the course, and profile-specific temporal patterns did not improve model fit. Instead, Deep Engagers showed consistently greater activation of mastery-, vicarious-learning-, and social-persuasion-aligned behaviors, especially around consequential assignments. The difference between the profiles may therefore lie less in the shape of their trajectories than in the intensity with which students mobilize these behaviors when the course becomes demanding.

The composite findings further suggest that self-efficacy is supported by multiple pathways. Students with more favorable behavioral, affective, and social profiles generally reported higher self-efficacy, but the relationship was not a steady dose-response pattern. Most of the difference occurred between students with no favorable profile and those with at least one. This may indicate that students do not need every aspect of their makerspace experience to be favorable. Instead, having at least one source of support (such as lower stress, deeper behavioral engagement, or stronger social participation) may provide an important foundation for confidence.

At the same time, the relationship between profiles and self-efficacy depended on students' sense of belonging. Among students with lower belonging, favorable profiles were clearly associated with higher self-efficacy. Among students with higher belonging, profile differences mattered very little. This pattern is consistent with belonging serving as a protective context. Students who felt connected to the makerspace community maintained relatively high self-efficacy even when their behavioral and affective profiles were otherwise less favorable.

Belonging may therefore reduce the extent to which confidence depends on students independently developing advantageous patterns of participation.

This finding extends prior work connecting joint visual attention, belonging, and self-efficacy by placing social participation within a broader configuration of students' makerspace experiences. Rather than treating belonging as another behavioral outcome, the present results suggest that it changes how behavioral and affective differences matter. A supportive community may not eliminate stress or make every student highly active, but it may change how students interpret those experiences and whether moments of difficulty undermine their confidence.

These findings have implications for makerspace facilitation. First, instructors should avoid relying on a single indicator of engagement. Students who appear active may still be struggling affectively, while students who spend considerable time observing may be participating productively. Second, observation, collaboration, and help-seeking should be designed as legitimate parts of making rather than treated as steps students must quickly leave behind. Third, facilitators may need to provide different forms of support to different students. Stress-Burdened students may benefit from help interpreting setbacks and regulating struggle, whereas Surface Engagers may benefit from clearer opportunities to move from observation into hands-on activity. Students who combine few favorable profiles with a weak sense of belonging may warrant particular attention.

More broadly, pose-based analytics may provide useful formative signals in environments where instructors cannot continuously observe every student. Their value, however, lies in complementing (and not replacing) surveys, conversations, and professional judgment. The profiles identified here should not be treated as fixed labels or automated diagnoses. They are contextual summaries of participation that may help facilitators notice patterns, ask better questions, and provide more responsive support.

5.1 Limitations

This study has several limitations that are worth mentioning. 1) The sample was small and came from three offerings of the same graduate course at one institution. The sample was also demographically narrow: participants were graduate students aged 25-35, and gender was imbalanced (48 women, 8 men), which limits the generalizability of the profiles to more diverse or younger populations and to non-graduate or non-academic makerspace settings. The profiles may therefore reflect this particular population, course structure, physical space, or set of tools.

They should not be assumed to generalize to other makerspaces without further study. 2) The profiles were also exploratory. Although we used theoretical criteria, stability checks, and cohort-balance tests, the clusters should not be treated as fixed student types. They are descriptive patterns of participation within this course. Because the profile search was partly guided by outcome associations, the links with self-efficacy should be interpreted cautiously and replicated in new samples. 3) The behavioral measures are proxies rather than direct measures of Bandura's sources of self-efficacy. Machine use may create opportunities for mastery, but it does not show that mastery occurred. Observation may support vicarious learning, but it may also reflect waiting. Joint visual attention and help-seeking are similarly open to multiple interpretations. The ONA pillars should therefore be understood as theory-aligned behavioral patterns, not direct measurements of psychological processes. 4) The affective profile relied on weekly self-reports and was driven mainly by stress. Semester averages may hide important changes around difficult assignments, deadlines, or project failures. In addition, the study used a general self-efficacy measure rather than a maker-specific scale. 5) Finally, the study was observational. It cannot establish whether belonging protects self-efficacy, whether self-efficacy increases belonging, or whether both develop together. Pose-based analytics also raise concerns about privacy, consent, and labeling. These profiles should not be used for automated decisions without strong contextual evidence and human judgment.

5.2 Future work

Future research should first examine whether the profile dimensions identified here replicate in larger and more diverse makerspace settings. Studies across institutions, age groups, course formats, physical layouts, and fabrication tools could clarify which patterns are robust and which depend on local features of the environment. Larger samples would also make it possible to use confirmatory profile methods, validate the exploratory profile structure on held-out data, and examine whether profiles differ across students' prior experience, identities, or educational backgrounds.

Future studies should also strengthen the interpretation of pose-derived behavioral indicators by combining them with richer measures of students' experiences. Interviews, stimulated-recall sessions, project logs, experience-sampling prompts, instructor observations, and records of successful or unsuccessful tool use could help distinguish productive observation from passive waiting, strategic help-seeking from unresolved difficulty, and mastery experiences

from repeated but unsuccessful interaction. Such data would allow researchers to test more directly whether the behavioral sequences identified through ONA correspond to the psychological processes described by Bandura.

Another further direction is to examine participation profiles as dynamic states rather than semester-long categories. Students may move between deeper and more peripheral engagement across assignments, tools, or moments of difficulty. Analyses conducted at the level of weeks, sessions, or consequential events could identify when these shifts occur and what features of the learning environment prompt them. This approach may be especially useful around deadlines, project transitions, technical breakdowns, and other moments when the differences between students become most visible.

The relationship between belonging and self-efficacy also deserves particular attention. Experimental or design-based studies could test whether structured peer support, mentoring, collaborative routines, public sharing, or changes in facilitation strengthen belonging and reduce the extent to which self-efficacy depends on students' behavioral or affective profiles. Repeated measures of belonging and self-efficacy would also help clarify their direction of influence and determine whether belonging precedes, follows, or develops alongside changes in participation.

Finally, future work could investigate how behavioral analytics can support instructors without turning exploratory patterns into fixed labels. Profile-informed systems might alert facilitators to students who are highly active but persistently stressed, or to students who remain peripheral across several sessions. These systems should be developed collaboratively with students and educators, provide interpretable evidence rather than categorical judgments, and be evaluated for privacy, fairness, and unintended consequences. The most useful role for pose-based analytics may be not to determine what a student needs, but to help instructors recognize when a closer conversation or observation may be warranted.

6. Conclusion

The broader implication of this work is that supporting students in makerspaces requires more than giving them access to tools and open-ended projects. Students need environments where observation, experimentation, collaboration, and help-seeking are treated as legitimate parts of learning. They also need support for managing the uncertainty and frustration that often accompany making. Instructors should therefore look beyond visible productivity and attend to

how students experience their participation, especially when sustained activity may be masking stress or low confidence.

The findings also highlight the importance of belonging. A strong sense of connection to the makerspace community may help students remain confident even when their patterns of participation are less favorable. This suggests that community-building should not be treated as an optional addition to technical instruction, but as a central part of makerspace design. Structured peer support, inclusive facilitation, and opportunities for students to contribute visibly to the community may be as important as access to machines or materials.

Finally, pose-based analytics may help instructors notice forms of participation that are difficult to track in busy, open-ended environments. Their most useful role is not to classify students, but to support reflection and more responsive teaching. Used carefully, these methods can help educators ask better questions about who is participating, who is struggling, and what kinds of support might help more students see themselves as capable makers.

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Highlights

- Pose-based video profiles reveal individual differences in makerspace engagement.
- Two stable profiles emerged: an affective and a behavioral dimension.
- Combining profiles predicted self-efficacy better than either alone.
- Students’ sense of belonging moderated this profile-efficacy relationship.
- Deep engagers showed stronger mastery, vicarious, and social network ties.